

SQL Handbook (301–350 Commands with Before & After Tables - Exact Model)

301 ALTER RENAME

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
ALTER TABLE students RENAME TO students_new;
```

After:

Table renamed

302 RENAME COLUMN

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
ALTER TABLE students RENAME COLUMN name TO full_name;
```

After:

Column renamed

303 ADD CONSTRAINT

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
ALTER TABLE students ADD CONSTRAINT chk_age CHECK(age>0);
```

After:

Constraint added

304 DROP CONSTRAINT

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
ALTER TABLE students DROP CHECK chk_age;
```

After:

Constraint removed

305 UNIQUE

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
ALTER TABLE students ADD UNIQUE(name);
```

After:

Unique added

306 PRIMARY KEY ADD

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
ALTER TABLE students ADD PRIMARY KEY(id);
```

After:

PK added

307 PRIMARY KEY DROP

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
ALTER TABLE students DROP PRIMARY KEY;
```

After:

PK removed

308 FOREIGN KEY ADD

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
ALTER TABLE students ADD FOREIGN KEY(id) REFERENCES dept(id);
```

After:

FK added

309 FOREIGN KEY DROP

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
ALTER TABLE students DROP FOREIGN KEY fk_id;
```

After:

FK removed

310 CHECK

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT * FROM students WHERE age>18;

After:
Filtered
```

311 DEFAULT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
ALTER TABLE students ALTER city SET DEFAULT 'Chennai';

After:
Default set
```

312 COMMENT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
ALTER TABLE students COMMENT='Student table';

After:
Comment added
```

313 ENGINE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
ALTER TABLE students ENGINE=InnoDB;

After:
Engine changed
```

314 CHARSET

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
ALTER TABLE students CONVERT TO CHARACTER SET utf8;

After:
```

Charset changed

315 COLLATE

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
ALTER TABLE students COLLATE utf8_general_ci;

After:
Collation set

316 RENAME INDEX

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
ALTER TABLE students RENAME INDEX idx_name TO idx_new;

After:
Index renamed

317 ADD INDEX

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
ALTER TABLE students ADD INDEX(age);

After:
Index added

318 DROP INDEX

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
ALTER TABLE students DROP INDEX age;

After:
Index dropped

319 SHOW INDEX

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SHOW INDEX FROM students;

After:
Index list

320 ANALYZE INDEX

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
ANALYZE TABLE students;

After:
Analyzed

321 CHECKSUM

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
CHECKSUM TABLE students;

After:
Checksum value

322 SHOW OPEN TABLES

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SHOW OPEN TABLES;

After:
Open tables

323 SHOW TABLE STATUS

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SHOW TABLE STATUS;

After:
Table info

324 SHOW TRIGGERS

Before Table (students):
| id | name | age | city |

```
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

SQL:
SHOW TRIGGERS;

After:
Triggers list

325 SHOW EVENTS

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SHOW EVENTS;

After:
Events list

326 SHOW PROCEDURE

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SHOW PROCEDURE STATUS;

After:
Procedures

327 SHOW FUNCTION

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SHOW FUNCTION STATUS;

After:
Functions

328 CALL PROCEDURE

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
CALL GetStudents();

After:
Procedure output

329 DECLARE VAR

```
Before Table (students):  
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:  
SET @a=10;
```

```
After:  
Variable set
```

330 SELECT VAR

```
Before Table (students):  
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:  
SELECT @a;
```

```
After:  
10
```

331 IF BLOCK

```
Before Table (students):  
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:  
SELECT IF(1=1, 'Yes', 'No');
```

```
After:  
Yes
```

332 LOOP

```
Before Table (students):  
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:  
SELECT 'Loop concept';
```

```
After:  
Loop
```

333 WHILE

```
Before Table (students):  
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:  
SELECT 'While concept';
```

```
After:
```

While

334 REPEAT

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT 'Repeat concept';

After:
Repeat

335 CURSOR

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT 'Cursor concept';

After:
Cursor

336 HANDLER

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT 'Handler concept';

After:
Handler

337 SIGNAL

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT 'Signal concept';

After:
Signal

338 RESIGNAL

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore


```
SQL:
SELECT 'Resignal concept';

After:
Resignal
```

339 SHOW VARIABLES LIKE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SHOW VARIABLES LIKE '%version%';

After:
Filtered vars
```

340 SET VARIABLE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SET sql_mode='';

After:
Mode set
```

341 SHOW WARNINGS LIMIT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SHOW WARNINGS LIMIT 1;

After:
Warning
```

342 SHOW ERRORS LIMIT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SHOW ERRORS LIMIT 1;

After:
Error
```

343 RESET SLAVE ALL

```
Before Table (students):
| id | name | age | city |
```

```
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
RESET SLAVE ALL;
```

After:

Reset done

344 SHOW GRANTS USER

Before Table (students):

```
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
SHOW GRANTS FOR CURRENT_USER;
```

After:

Grants

345 SHOW STATUS LIKE

Before Table (students):

```
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
SHOW STATUS LIKE 'Threads%';
```

After:

Status

346 SHOW ENGINE STATUS

Before Table (students):

```
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
SHOW ENGINE INNODB STATUS;
```

After:

Engine info

347 SHOW FULL TABLES

Before Table (students):

```
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
SHOW FULL TABLES;
```

After:

Tables

348 SHOW PLUGINS

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SHOW PLUGINS;

After:
Plugins

349 SHOW PRIVILEGES

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SHOW PRIVILEGES;

After:
Privileges

350 SHOW PROCESSLIST FULL

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SHOW FULL PROCESSLIST;

After:
Processes